

Experiment 6: Champagne Tower

Aim: To determine how much champagne is present in a particular glass of a champagne tower after pouring a given amount of champagne into the top glass.

Algorithm

1. Read the amount of champagne poured, the query row, and the query glass.
2. Initialize the top glass with the poured amount.
3. If a glass contains more than one cup, keep one cup in it and distribute the excess equally between the two glasses below it.
4. Repeat this process row by row.
5. Find the amount in the required glass.
6. Print the result, with no glass containing more than one cup.

Python Program

```
def champagneTower(poured, query_row, query_glass):  
    tower = [[0.0] * (query_row + 1) for _ in range(query_row + 1)]  
    tower[0][0] = poured  
    for i in range(query_row):  
        for j in range(i + 1):  
            if tower[i][j] > 1:  
                excess = tower[i][j] - 1  
                tower[i + 1][j] += excess / 2  
                tower[i + 1][j + 1] += excess / 2  
    return min(1.0, tower[query_row][query_glass])  
  
poured = float(input("Enter amount of champagne poured: "))  
query_row = int(input("Enter query row: "))  
query_glass = int(input("Enter query glass: "))  
result = champagneTower(poured, query_row, query_glass)  
print("Amount of champagne in glass:", result)
```

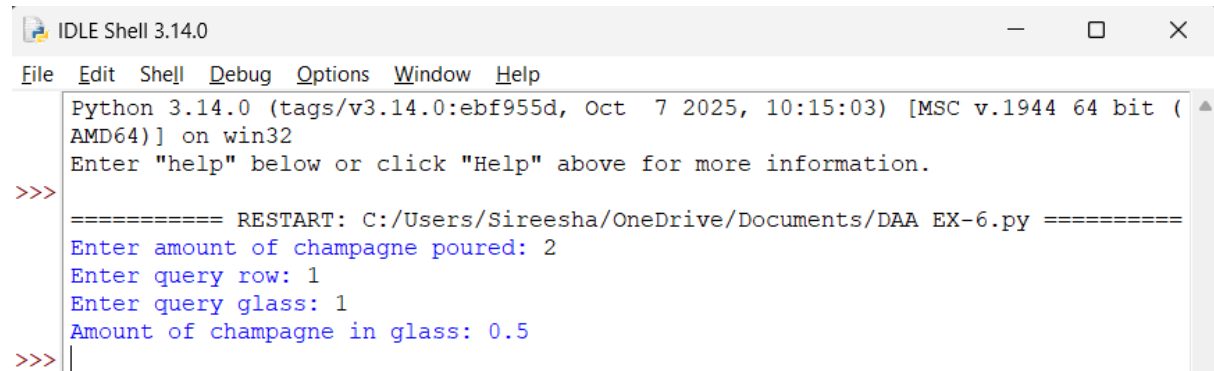
Input

Enter amount of champagne poured: 2

Enter query row: 1

Enter query glass: 1

Output



```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==== RESTART: C:/Users/Sireesha/OneDrive/Documents/DAA EX-6.py =====
Enter amount of champagne poured: 2
Enter query row: 1
Enter query glass: 1
Amount of champagne in glass: 0.5
>>>|
```

Result

Thus, the Python program to calculate the amount of champagne in a specified glass of the champagne tower was successfully executed.